

Department of Computer Science and Engineering
Sejong University
Signals and Systems

Course # 005246

Quiz # 1

Credit Hours: 03
Quiz Date: April 6, 2026
Course Instructor: Prof. Amad Zafar

Semester: 2026-I
Total Marks: 10
Quiz Time: 15 Minutes

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| <p>1) The analog signals are</p> <ul style="list-style-type: none">☒ a) Continuous in time and continuous in valueb) Continuous in time and discrete in valuec) Discrete in time and continuous in valued) Discrete in time and discrete in value <p>2) A function $f(t)$ is an even function if for all values of t</p> <ul style="list-style-type: none">a) $f(t) = -f(t)$☒ b) $f(-t) = f(t)$c) $f(-t) = -f(-t)$d) None of the above <p>3) When a unit step function is multiplied by a non-causal signal, the resultant signal will be</p> <ul style="list-style-type: none">a) non-causalb) anti-causal☒ c) causald) None of the above <p>4) The function $\delta[2n]$ is equal to</p> <ul style="list-style-type: none">☒ a) $\delta[n]$b) $2\delta[n]$c) $\frac{1}{2}\delta[n]$d) $\delta[\frac{1}{2}n]$ <p>5) In the unit impulse function, the unit term represents.</p> <ul style="list-style-type: none">a) Height is 1☒ b) Area is 1c) Duration is 1d) None of the above <p>6) Adding a DC value to a periodic signal will change the periodicity of the signal.</p> <ul style="list-style-type: none">a) Yes☒ b) No | <p>7) Power signals are the signals with</p> <ul style="list-style-type: none">a) $0 < E < \infty, P = 0$b) $0 < P < \infty, E = 0$☒ c) $0 < P < \infty, E = \infty$d) $0 < E < \infty, P = \infty$ <p>8) The fundamental period of $\cos^2 t$ is.</p> <ul style="list-style-type: none">☒ a) πb) $\pi/2$c) 2πd) $\pi/4$ <p>9) The fundamental period of $x[n] = (-1)^n$ is.</p> <ul style="list-style-type: none">a) 4☒ b) 2c) πd) 1 <p>10) The system $y(t) = x(t) + \cos(t)$ is</p> <ul style="list-style-type: none">a) Linear and time-invariantb) Linear and time-varying☒ c) Non-linear and time-varyingd) Non-linear and time-invariant |
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